Dániel Szekeres

+36 (20) 278 8443 • szekeresdani@gmail.com • szdan97
dániel-szekeres-0619b8178 • 0000-0002-2912-028X

Experience

Budapest University of Technology and Economics (BME) **2018–present**
Critical Systems Research Group (FTSRG)
Assistant Research Fellow since February 2026; before that PhD student (2022–2026) and BSc/MSc student (2018–2022), with teaching and research assistant contracts throughout.

- Member of the core developer team of Theta, an open-source model checking framework (Java/Kotlin) with entries in the SV-COMP software verification competition (2024, 2025); designed and built its probabilistic extension (GitHub).
- Worked on every layer of Theta, from formalism frontends to debugging the SMT interface, with most of the work on the verification algorithms.
- Coordinated Theta-related BSc and MSc student projects (typically three supervised students at a time): shaped each project to advance a shared module, reviewed designs and code in weekly consultations, and aligned the work with that of the other core developers, generalizing features for reuse. Topics included concurrency, partial order reduction, timed systems, hardware model checking and probabilistic model checking.
- Co-supervised nine papers at the faculty's student research conference (2022–2025), all awarded: four first and five second prizes.
- Industrial projects: reliability analysis of a railway signaling architecture; formal verification in the architecture design of a fault-tolerant steer-by-wire system.
- Teaching: formal methods, automated verification (with labs on static analysis, Z3 and implementing model checking algorithms), probability and statistics, systems engineering, system modeling.

Prolan Irányítástechnika Zrt. **Summer 2019**
Intern, Reliability Analysis
Quantitative reliability analysis of a SIL4 railway signaling system.

BME Formula Racing Team (driverless) **2017–2018**
Environment Perception Engineer
Worked on SLAM, then built the LiDAR- and camera-based perception subsystem (C++) of the driverless car. A small onboarding project involved reinforcement learning.

Research

Software model checking. Abstraction-based verification of C programs in Theta, especially concurrent programs (happens-before reasoning, on-the-fly cone-of-influence reduction).
Co-author: *SPIN 2025*, *TACAS 2025* (reproducibility study), *DSN Workshops 2025*; four SV-COMP competition contributions (*TACAS 2024*, *2025*).

Distinct requirement violation scenarios. Human-in-the-loop process that makes model checkers return only genuinely distinct counterexamples; applied to an automotive steer-by-wire design.
R. Szabó, D. Szekeres, et al. *Empirical Software Engineering* 31(3), 2026.

Lazy abstraction of Markov decision processes. First lazy abstraction algorithm for MDPs; exact and approximate variants, combined with Bounded Real-Time Dynamic Programming.
D. Szekeres, K. Marussy, I. Majzik. *ASMTA 2024*; extended version under review at *ACM Transactions on Computational Logic*.

Abstraction-based analysis of probabilistic programs. Extended game-based abstraction refinement to probabilistic C programs with new abstract domains and encodings.
D. Szekeres, I. Majzik. *Acta Cybernetica* 26(3), 2023.

Tensor-based reliability analysis. Reliability analysis of large fault trees with Tensor Trains, avoiding explicit state-space construction.
D. Szekeres, K. Marussy, I. Majzik. *EDCC 2021*.

16 peer-reviewed publications in total; full list via ORCID.

Education

BME **2022–2026**
PhD in Computer Science
Dissertation: *Efficient Reliability Analysis of Complex Critical Systems*. Studies completed January 2026; internal defense completed, public defense pending (expected in March 2027).

BME **2020–2022**
MSc in Computer Science Engineering, with honors (5.0 GPA)
Critical Systems major, Intelligent Systems (AI) minor.

BME **2016–2020**
BSc in Computer Science Engineering, with honors (5.0 GPA)
Systems Engineering specialization.

AI/ML: Artificial Intelligence; Neural Networks; Deep Learning in Practice; Machine Learning; Probabilistic Inference and Decision Support Systems; Security in Machine Learning; Complex Federated Models in Machine Learning; Distributed Intelligent Systems (multi-agent systems). Coursework used Keras, TensorFlow, PyTorch; self-study of reinforcement learning (Sutton & Barto).

Skills

Formal methods: Model checking, SMT-based verification, program semantics, specification and temporal logic, concurrency, probabilistic systems and Markov decision processes, counterexample analysis

Programming: Java and Kotlin (daily); Python; C, C++; learning Lean and Rust; coursework in Verilog, Erlang, Prolog, JavaScript

Engineering: Git-based collaboration, continuous integration, testing, code review, Docker

Languages: Hungarian (native), English (C1), German (B2)